Sequence Pattern => Sekuensial

↳ Terdiri atas ordered elements atau items

Transaction database vs Sequential database.

a, b, c <a, (abc)(ac) d (cf)>

a c d <(ad) c (bc) (ae)>

~~Apl~~ Application of sequential pattern mining :

- Medical treatment — disease conditions
- Natural disaster — Weblog click streams
- DNA sequence and gene structures.

Sequence => ordered list of events : $\langle e_1, e_2, \ldots, e_n \rangle$

α merupakan subsequence β, $\alpha \subseteq \beta$

$\langle a(bc)dc \rangle$

Sequential Pattern Mining
- → Apriori-based approaches.
  - GSP
  - SPADE
- → Pattern-Growth based approaches.
  - Free Span
  - Prefix Span

$$\text{Confidence } A \to B = \frac{\text{Support}(A,B)}{\text{Support}(A)}$$

$$\text{Support}(A,B) = \frac{\text{Frequency}(A,B)}{N}$$

Association rule => Suatu pola yang menggambarkan suatu item (atau event) yang ~~terasosiasi~~ sering terasosiasi bersama (frequently associated together).

- → : → Support : dari berapa kejadian, berapa kali terjadi
  - ↳ Confidence : dari satu kejadian terjadi, berapa ~~kepastiannya~~ keyakinannya.
- → memiliki minimum support threshold dan minimum confidence ~~support~~ threshold

$$\text{Support}(A,B) \Rightarrow \frac{\text{Frequency}(A,B)}{N}$$

$$\text{Confidence}(A \Rightarrow B) \Rightarrow \frac{\text{Support}(A,B)}{\text{Frequency}(A)} = \frac{2}{4} = 50\%$$

$$\frac{\text{Frequency}(A)}{N} \propto \frac{1}{\text{Frequency}(A)}$$

$$\frac{2}{4} \times \frac{N}{3/4} = \frac{2}{4} \times \frac{4}{3} = 66{,}67\%$$

$$\frac{2}{4} \times \frac{1}{2/4} = \frac{2}{4} \times \frac{4}{2} = 100\%$$

$$\text{Confidence}(A \Rightarrow B) = P(B|A) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

Definitions

- Itemset : a set of items ~~itemset~~ (min 1 element)
- Frequency : Jumlah transaksi yang mengandung itemset.
- Frequent itemset : Itemset yang memenuhi minimum support.

③ Two steps of frequent pattern:

Find all frequent itemset => Apriori algorithm

Generate strong association rule => Satisfy minimum support and minimum confidence.

Apriori algorithm

Any subset of a frequent itemset must be frequent.
  # if {milk, bread, nuts} is frequent, so is {milk, bread}

Apriori pruning principle

Jika ada itemset yang tidak sering muncul (infrequent), maka supersetnya tidak bisa dibangkitkan (diuji).

Method

- Generate length (k+1) candidate itemset from length k frequent itemset, and
- Test the candidates against DB.

Find the strong association rules:

- All frequent itemsets satisfy minimum support.
- Get the ones that satisfy the minimum confidence.

Contoh penerapan FP:

Market basket analysis => Memprediksi kemungkinan membeli barang lain apabila membeli suatu barang tertentu.

Kecenderungan pengambilan matakuliah.

Association Rule

Association rule = $(m-p) \rightarrow p$

- Confidence = $\frac{sup(m)}{sup(m-p)}$
- Support = $sup(m)$

Tantangan Frequent Pattern Mining:

- Multiple scan database. (Pindai database berulang kali).
- Huge number candidates (Jumlah kandidat yang dihasilkan besar).
- Pelurusan yang membosankan.

Frequent Pattern Growth (FP-Growth).

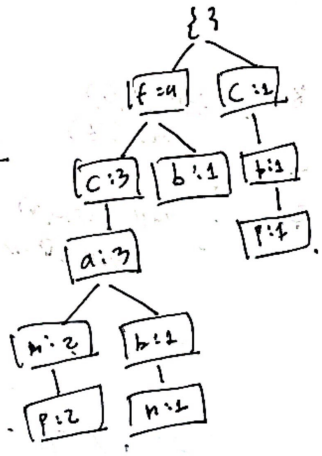
~~Fre~~ Adopting the divide-and-conquer approach.
  ↳ ~~FP-tree~~ Pertama, bentuk FP-tree.
  ↳ ~~Ke~~ bentuk conditional database.

Method
  - Bentuk conditional pattern base, kemudian conditional FP-tree
  - Ulangi proses sampai setiap conditional FP-tree terbentuk.
  - Sampai terbentuk FP-tree kosong atau hanya mengandung satu path, satu path akan menghasilkan berbagai yang menjadi sub path-nya.

⑤ Construct FP-tree

1. Scan DB once, find frequent 1-itemset (single item pattern).
2. Sort frequent items in frequency descending order, f-list
3. Scan DB again, construct FP-tree.

→ Lihat min-support-nya!



Conditional Pattern-base

Item dengan min support	Conditional pattern base	Conditional FP tree yang memenuhi min support=3
p	{fcam:2}, {cb:1}	{c:3}
m	{fca:2}, {fcab:1}	{fca:3}
b		
a	{fc:3}	{fc:3}
c	{f:3}	{fc:3}
f		

Bentuk frequent dengan menggunakan conditional FP-tree yang memenuhi minimum support dengan itemset yang memenuhi minimum support.

{c:3} × {p} = ~~{fc:3}~~ {cp:3}

{fca:3} × {m} = {fm:3}, {cm:3}, {am:3}, {fcm:3}, ...

⑥ Benefit of the FP-tree structure

- Completeness
- Compactness

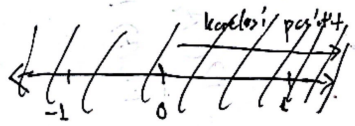
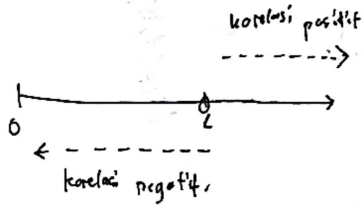
Pattern evaluation ⇒ using correlation analysis

Correlation rules:

A ⇒ B [support, confidence, correlation]

$$\boxed{Lift(A,B) = \frac{P(A \cup B)}{P(A)\,P(B)}}$$

< 1 : korelasi negatif

> 1 : korelasi positif

1 : Variabel saling independen. ⇒ Tidak ada korelasi